Sergii Kryvonos

EDUCATION:
2003 - 2009: National Technical University “KhPI”
Computer and Information Technologies Facility – System Programming
Engineer System Analytic (Major: System Programming).
Relevant courses included system programming (c/c++, assembler), databases, applied and computing mathematics, computer modelling, computer graphics.

CORE SKILLS:
Languages:
C/C++, C#, ASP .NET, PHP, Perl, Python, Java, Kotlin, Prolog
OS:
FreeBSD, MacOS, Haiku
Debian/Ubuntu/RHEL/Fedora/CentOS/Oracle/Suse/Gentoo Linux, Solaris
Android, iOS, WindowsPhone
Windows family, Mobile;
Arduino, Raspberry PI, Pi Zero, Rockchip, WebOS
DBMS:
MSSQL, MySQL/MariaDB, PostgreSQL, Oracle, SQLite, Access
IDE:
Eclipse, NetBeans, MS Visual Studio, Microsoft Expression Blend, #develop, KDevelop 3, 4, QtCreator,
Code::Blocks, MonoDevelop, Xcode, VScode
Zend Development Environment, Borland C++ Builder, Delphi
Office Software:
MS Office, Open Office, MS Project, PowerPoint, Keynote
Tools:
IDA, CMake, Conan, Automake, Virtual BOX, qemu
Other tools, technologies:
Office solutions development, deployment systems/installation packages development, WIX, regular expressions, reverse engineering.
API hooking, injection, xperf profiler, Apple profilers, Visual Studio profilers, Valgrind
OOP, STL, boost, ACE, MFC, Qt
UWP, .NET 2.0, 3.5, 4.5, 5.0. WPF, WCF, Silverlight, .NET 2.0 CF, Windows Forms, GDI+
HTML, CSS, Direct3D, Ogre3D, OpenGL ES2, Barcode recognizing, Porting Java to .NET and .NET to C++
bash, batch, various bug-trackers, project support systems, continuous integration
GIT, Perforce, TFS, Subversion (SVN), CVS, SC
Gerrit, Gitlab, CodeCollab, GitHub, Atlassian
LANGUAGES:
English – Fluent
German - A2
Ukrainian – Native
Russian – Native
UNIQUE EXPERTISE:
Dynamic Programming using own STM R&D, abstract math expressions with calculations caching (NoSQL DBs), generated OpenCL kernels calculated graphics, etc.
Newest C++ features ideas authoring (ready to become part of C++ working group).

CAREER HISTORY:

Miltenyi Biotech, Bergisch Gladbach, Germany, since 04.2025.
Cythometry/separation embedded code scanner device integration.
Contributed into C++ platform, cmake/vcpkg build system, presets, Yocto-based OS.
Driving distributed compilation practice, researched required ports, communicated with IT.
Driving WSL2 embedded practice: image import, Cuda, USB passthrough.
Introduced CMake presets to platform.

Apex Tool Group, Westhausen, Germany, 05.2024 – 09.2024
Embedded C++ supporting multiple embedded vcpkg components
Kotlin support UI Windows application, and installer
EEPROM write journaling player
Logging components
iBeacon daemon implementation
connection stability client-server statistics gathering
POC VNC service deployment with bash and SSH

Qt, Berlin, Germany, 09.2023 – 01.2024
Cross-platform, Embedded
VsCode plugin maintainer
Implemented new VScode plugin with CMake/QMake support from scratch.
Three months from zero to actual working prototypepresented at the Qt contributors summit, may
be a record.
Trusted with maintainership role from the beginning
VScode extension design and implement from scratch
(https://codereview.qt-project.org/q/project:qt-labs%252Fvscodeext+status:open)
Implemented C++ support from scratch for QMake and CMake build Qt applications.
QtCreator and other Qt contributions
(https://codereview.qt-project.org/q/author:serg.kryvonos%2540qt.io)
sed-fix for lambdas batch fix and other C++20 compatibility fixes.

SoftServe, Lviv-Gdańsk, Ukraine-Poland, 10.2022 – 05.2023
Cross-platform, Embedded
NDA new product device (Logitech)
POC prototyping and R&D on new kind of device.
Technologies: C/C++, OpenCL, OpenCV, CMake, Visual Studio/Xcode, git

SantaBrowser, 06.2022 - 10.2022
C++ Independent Contractor
Chromium-based proprietary “Crypto”-browser
CI for the Browser based on GitHub, Jenkins and Docker
Implement and upstream required changesUse macOS-compatible shebang (3810246) · Gerrit Code
Review(https://chromium-review.googlesource.com/c/devtools/devtools-frontend/+/3810246)
Technologies: C++, npm, gn, Gerrit, GitHub, Jenkins

Dash Financial Technologies,01.2021 - 05.2022
C++ Independent Contractor
ULL
Low-latency trading platform
Technologies: C++, CMake, Conan

MariaDB, 10.2021 - 12.2021
Contributor
Server (https://github.com/MariaDB/server)
Optimizer traces JSON consistency fixes.
Technologies: C++, JSON, BuildBot, Linux, cmake, perl, macOS, QtCreator, Eclipse CDT, git, github.

Svitla, 04.2020 - 04.2021
Technical LeadSDE
LibraryFiles (Thermofisher Scientific)
Genetics microarray designer configuring tool.
Technologies: C++, XML, JSON, Jenkins, PostgreSQL, Boost, Gtest, boost unittest framework.

GlobalLogic, 05.2019 – 04.2020
Technical LeadSDE
Incredibuild
Debugging and support for build acceleration tool.
Technologies: C++, Delphi, usermode virtualization, process offloading.

SoftServe, Kharkiv, Ukraine, 09.2017 – 11.2018
Cross-platform, EmbeddedSDE – Abiliton TechnicalLead
Meet Up Camera (Logitech)
Detection library software development assist.
Technologies: C/C++, OpenCL, OpenCV, CMake, Visual Studio/Clion/Xcode/Kdevelop/ Eclipse, git
CorelDraw/PhotoPaint (Corel)
Windows/MacOS C++ development.
Technologies: C/C++, Git/Gerrit, Visual Studio/Eclipse/Xcode

RedPointPositioning, Kharkiv, Ukraine, 5.2017 - 7.2017
EmbeddedSDE
message processing component
Translate DB queries into REST API calls.
Technologies: C, Makefile, Eclipse, Amazon

Querix, Kharkiv, Ukraine, 1.2017 – 4.2017
C++ SDE
TNS protocol developer
Stabilise TNSrun oracle db connection client. Reimplement network layer to boost::asio
Technologies: C/C++, STL, boost, asio, VisualStudio, git, Linux, Oracle DBMS, MSSQL

DasPlast, Kharkiv, Ukraine, 07.2015 – 12.2016
Voluntary startup investment and consulting
Firmware Development
RAMPS customisation.

Technologies: C/C++, RAMPS, 3dPriner, Linux, Eclipse, git
Website Development
OpenShift, PHP, CSS, JavaScript, OpenCart.
Technologies:OpenShift, PHP, CSS, JavaScript, OpenCart,Eclipse, PHP Storm, git

Luxoft, Kiev, Ukraine, 07.2012 – 07.2015
Technical Lead
WebOS Development (LG)
Supporting OpenWebOS and WebOS Pro.
Technologies: C/C++, Python, openembedded, Linux, Eclipse, Bitbake, CMake, git
Kernel mode driver development (AMD)
Work in 20+ people project concentrating on providing components of fundamental interfaces to the
Windows operating system, mostly in kernel space, some are in user space. Driver implementation based
Microsoft WDDM specification, which supports Windows Vista, 7 and 8. Customer is one of the biggest
semiconductor companies.
Technologies: C++, Microsoft WDDM, Windows Vista, 7, 8, Eclipse, automake

FaceUpGaming (http://www.gotfacegames.com), Kharkiv, Ukraine 01-07/2012
Technical Lead, IT Director in Kharkiv office
Poker Backend
Poker game server.
Responsibility — Interviewing/testing core people, Backend support, forming office staff, task management, coaching.
C++, CMake, sockets, STL, boost, Eclipse, git, Wireshark, Ubuntu Server, Gentoo Linux, Rackspace, XEN virtual machines, C#, mono, MySQL, MSSQL, etc.

SphereInc (http://www.sphereinc.com/), Kharkiv, Ukraine 02-11/2011
CIMS
CRM for medical industry including various issues management.
Responsibility — C# .NET 4 development.
Used technologies: MSVS 10, C# .NET 4, MVC2, Fluent, Nhibernate, jQuery.
BrightGaugeAgent
Provides secure access to local RMM and PSA data.
Responsibility — Python development, WIX installer creation, custom actions development in C#.
Usedtechnologies:Eclipse,Python,py2exe,SharpDevelop,WIX,C#.NET2.0,MySQL,MSSQL, SSL,JSON,
XML.
MenuDNA
Provides flexible solutions for restaurants.
Responsibility – iPhone/iPad development, estimate and develop solution.
Used technologies: Xcode 4, Objective C, JSON.
Herefirst
Provides list of nearby restaurants.
Responsibility – estimate and development for iPhone.
Used technologies: Xcode, Objective C, ZXing library.
Buddha(http://www.buddhatheapp.com/)
Buddha helps you expand your social network and find new connections based on your personal likes, interests, and location.

Responsibility - lead development of an Android version.
Used technologies: Eclipse, Java, ADT.

QArea (http://www.qarea.com/), Kharkiv, Ukraine 01/2010-08/2010
Senior Software Developer
MobileDocs
Mobile application for British corporation. Responsibility – C# development for Windows Mobile.
Used technologies: C# Windows Forms, scalable design and UI, CAB Setup.
Progamming in C# CF 2.0, C++ for Windows Mobile ARM platform.
Porting Zxing barcode recognizing library for CF 2.0.
Used technologies: CF 2.0, Zxing barcode recognizer, Capture frame bitmaps from device camera;
Using program database; Objects mapping as datasource.
GreenPerks (http://www.laketran.com/aboutus/GreenPerks/)
Mobile client for GreenPerks resource. Ordered by T-Mobile.
Responsibility – C# development for Windows Mobile, Porting from Android.
Used technologies: C# Windows Forms, scalable design and UI, Registry, CAB Setup.
Skyhook (http://skyhookwireless.com/)
GPS via WIFI/GSM.
Responsibility – C++ development, Python development.
Used technologies: STL, CMake, MSVC&GCC cross-platform code and unit tests.

Akvelon (http://www.akvelon.com/), Kharkiv, Ukraine 05/2009 – 12/2009
Team Lead/Tech Lead
DeskStream (ex NomaDrive, currentlyWorldDesk)
Windows Desktop virtualization.
Responsibility – C++ development, Team Lead, WINE development (C language).
Usedtechnologies: injecting, hooking, OOP, design patterns, STL, boost, unittests, interprocess data sharing, Eclipse, Automake.

Ambercore Software (4D-iQ SightPower http://sight-power.com/), Kharkiv, Ukraine 11/2008-04/2009
Senior Software Development Engineer
Ambercore Spatial Decisions Software Platform 2d/3d visualization engine.
Responsibility – 2d/3d graphical visualization engine support and development.
Used technologies: OOP, design patterns, STL, boost, GDI+, Direct3D, Ogre3D, .NET Framework 3.5, mixed managed/native C++.

Tech Data Solutions (http://www.techdatasolutions.com/ rebranded into Akvelon), Kharkiv, Ukraine 06/2007 - 10/2008
NomaDrive
Windows Desktop virtualization.
Responsibility – C++ development.
Used technologies: injecting, hooking, OOP, design patterns, STL, boost, inter process data sharing.
TProcess
Process based portal based on SharePoint for project workflow automation.
Responsibility – workflow editor development.
Used technologies: C#, .NET Framework 3.5, WPF
Development Environment: MS Visual Studio .NET 2005; SVN

Flexaspect (http://www.flexaspect.com/), Kharkiv, Ukraine 10/2006 – 06/2007
Postacrime project(www.postacrime.com)
Responsibility – site development (administrative and user area), scheduled parsers development.
Used technologies:PHP 5, Smarty-Based Object Oriented Engine, Linux Abilities (Cron, Wget, Gunzip), regular expressions.
Development Environment: ZDE
Komarovskiy(www.komarovsky.net)
Komarovskiy is resource devoted to medicine theme.
Responsibility – PHP development
Used technologies: PHP 5, Joomla CMS, phpBB for Joomla
Development Environment: ZDE
Database: MySQL
UkrDealing project(http://udinform.com/)
Responsibility – PHP development, Charts Engine
Used technologies: PHP 5, GD2, Joomla CMS
Development Environment: ZDE
Database: MySQL

OTHER PROJECTS:

OpenMind https://github.com/ohhmm/openmind
Responsibility – project authorship, design and development.
Used technologies: CMake,C++, boost, JetBrains CLion, GoalGraph, NeuralNetworks, abstract Math expressions and arbitrary system solver, boost::mpl
Development Environment: JetBrains CLion, Xcode, Eclipse, Visual Studio, Kdevelop

MyStride
Responsibility – horses gait detection on iPhone.
Used technologies: iOS, Objective C, PHP, C++, extrapolation.
Development Environment: Github, Xcode, Eclipse.

Nanny (see https://www.microsoft.com/en-us/store/p/nanny/9nblggh4sd84
and https://specialmeaning.blogspot.com/2016/09/nanny.html)
Responsibility - ownership, authorship, software development and market publishing.
Used technologies: UWP .NET for Microsoft Windows.
Development Environment: Microsoft Visual Studio.

SEM Journals (https://github.com/ohhmm/SEM_Journals_NET)
Responsibility – development.
Used technologies: UWP, ASP .NET, MVC, EntityFramework, WCF, WPF, .NET 4.6 (C# 7.0).
Development Environment: Microsoft Visual Studio.
FreeCAD (pull requests)
Responsibility – voluntary debugging.
Used technologies: CMake, C++, boost
Development Environment: Eclipse

Synergy through USB (https://github.com/ohhmm/synergy-through-usb)
Forked synergy and implemented USB link cable transport layer for secure sharing of mouse and keyboard between computers.
Used technologies: cmake, git-svn, c++, libusb, Qt, nsis.
Development Environment: Github, cmake, Eclipse, VS 2010.

OI-sync (https://github.com/iHateInventNames/shoppinglist)
Responsibility – forked openintents shopping lists to implement auto synchronization.
Used technologies: Android, Java.
Development Environment: Eclipse, git, github

OpenNURBS to IGES convertor for CAD formats
Responsibility – C++ programming, engineering convertor.
Used technologies: OpenCASCADE, OpenNURBS.
Development Environment: VS2008.

Vcxproj2cmake (https://github.com/envi/vcxproj2cmake)
Responsibility – Implemented VisualStudio Filters support and automatic scanning for library dependencies to include into generated CMakeLists.txt file.
Used technologies: Perl, CMake.
Development Environment: Gentoo Linux, Eclipse EPIC plug-in, CDT, VS Express.

RADIUS-protocol authentication service
Responsibility – Implement C++, Perl
Used technologies: Linux, perl, c++, ACE, boost, mysql, libradius, mysql++.
Development Environment: Ubuntu Linux, KDevelop3.

Registry ease-use module
Responsibility – C++ programming, engineering ease to use layer for MS Windows registry operations.
Used technologies: WinApi, optimized light-weight code.
Development Environment: VS6.

OpenGL ES2 Engine
Responsibility – C++ development, shaders, resource manager, textures, camera, etc.
Used technologies: WinApi, OpenGL
Development Environment: Visual Studio 2017
OpenGL Life game
Responsibility – engineering class map, physics, code (C++).
Used technologies: WinApi, OpenGL
Development Environment: VS8 (2005)

Cars (https://github.com/ohhmm/Cars)
Responsibility – implemented car racing game as part of interview.
Used technologies: C++, Irrlicht.
Development Environment: VS, git, GitHub

Micro Application Solution
Assembler development, processor pipelines optimization.
Used technologies: WinApi, MASM, speed optimization based on processor pipelines system knowledge.
Development Environment: MASM, AsmEdit

Marlin for SUMPOD and XY 3D Printers
Responsibility – adaptation of firmware for SUMPOD Aluminum and XY 3d printers.
Used technologies: Arduino, RAMPS, C++.
Development Environment: Eclipse, git, Arduino IDE

AI R&D
Published RELU squashing optimisation idea based on arbitrary precision
https://www.youtube.com/watch?v=WhkeXGapxzA&list=PLnI9xbPdZUAmUL8htSl6vToPQRRN3hhFp&index=10
https://github.com/ohhmm/openmind